

ГУАП

КАФЕДРА № 43

ОТЧЕТ
ЗАЩИЩЕН С ОЦЕНКОЙ
ПРЕПОДАВАТЕЛЬ

Должность

ассистент

подпись, дата

Кочин К.А

инициалы, фамилия

ОТЧЕТ О ЛАБОРАТОРНОЙ РАБОТЕ №4

Маршрутизация

по дисциплине: Администрирование вычислительных сетей

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Санкт-Петербург
2025

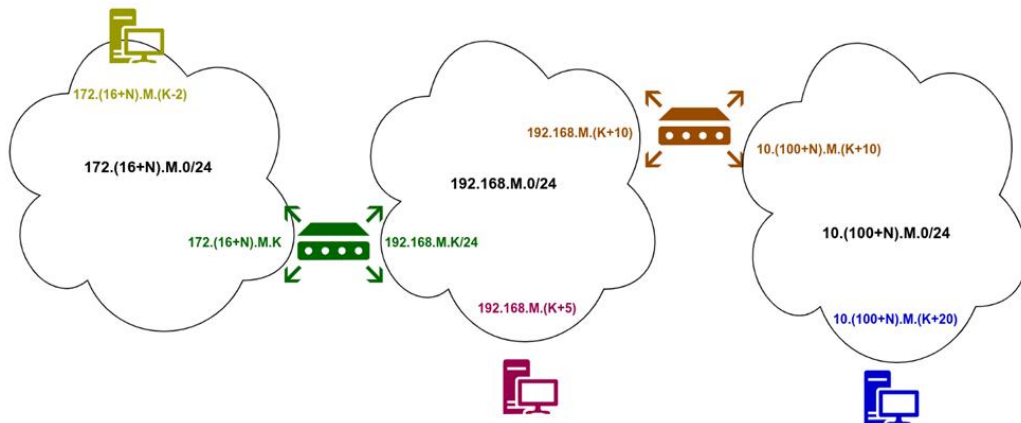
1. Цель Работы

Изучить теорию и практику маршрутизации пакетов и построения маршрутизируемых IP-сетей.

2. Задание

Необходимо:

1) Создать сеть из трех подсетей, содержащую 2 роутера и 3 клиента, как показано на рисунке ниже.



где

- N - последняя цифра из номера группы,
- M - ваш порядковый номер по журналу,
- K - длина вашей фамилии.

Клиентские машины должны иметь один сетевой интерфейс.

Роутеры – два.

- 2) Настроить адреса всех интерфейсов в соответствии с вариантом.
- 3) Настроить таблицы маршрутизации, чтобы все клиенты могли общаться друг с другом.
- 4) Пропинговать с каждого клиента двух остальных и зафиксировать результат.

3. Скриншоты или вывод из консоли с результатами работы и соответствующими комментариями;

Для ClientA

ping -c 3 192.168.15.10 # ClientB

ping -c 3 10.106.15.10 # ClientC

ping -c 3 192.168.15.1 # Router1 (local)

ping -c 3 192.168.15.2 # Router2 (through Router1)

```
main@clientA:~$ ip route
default via 172.22.15.1 dev enp0s8
172.22.15.0/24 dev enp0s8 proto kernel scope link src 172.22.15.10 metric 100
main@clientA:~$ ping -c 3 192.168.15.10
PING 192.168.15.10 (192.168.15.10) 56(84) bytes of data.
64 bytes from 192.168.15.10: icmp_seq=1 ttl=63 time=13.3 ms
64 bytes from 192.168.15.10: icmp_seq=2 ttl=63 time=3.13 ms
64 bytes from 192.168.15.10: icmp_seq=3 ttl=63 time=3.44 ms

--- 192.168.15.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2008ms
rtt min/avg/max/mdev = 3.125/6.626/13.313/4.729 ms
main@clientA:~$ ping -c 3 10.106.15.10
PING 10.106.15.10 (10.106.15.10) 56(84) bytes of data.
64 bytes from 10.106.15.10: icmp_seq=1 ttl=62 time=3.54 ms
64 bytes from 10.106.15.10: icmp_seq=2 ttl=62 time=2.67 ms
64 bytes from 10.106.15.10: icmp_seq=3 ttl=62 time=2.84 ms

--- 10.106.15.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2008ms
rtt min/avg/max/mdev = 2.670/3.016/3.535/0.373 ms
main@clientA:~$ ping -c 2 192.168.15.1
PING 192.168.15.1 (192.168.15.1) 56(84) bytes of data.
64 bytes from 192.168.15.1: icmp_seq=1 ttl=64 time=28.5 ms
64 bytes from 192.168.15.1: icmp_seq=2 ttl=64 time=2.75 ms

--- 192.168.15.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1004ms
rtt min/avg/max/mdev = 2.750/15.637/28.524/12.887 ms
main@clientA:~$ ping -c 2 192.168.15.2
PING 192.168.15.2 (192.168.15.2) 56(84) bytes of data.
64 bytes from 192.168.15.2: icmp_seq=1 ttl=63 time=2.68 ms
64 bytes from 192.168.15.2: icmp_seq=2 ttl=63 time=1.44 ms

--- 192.168.15.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.442/2.060/2.678/0.618 ms
main@clientA:~$
```

Для clientB

ping -c 2 172.22.15.10 # ClientA

ping -c 2 10.106.15.10 # ClientC

ping -c 2 172.22.15.1 # Router1

ping -c 2 192.168.15.2 # Router2

```
main@clientB: ~  
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr  
oup default qlen 1000  
    link/ether 08:00:27:85:29:25 brd ff:ff:ff:ff:ff:ff  
    altnam enx080027852925  
    inet 192.168.15.10/24 brd 192.168.15.255 scope global noprefixroute enp0s8  
        valid lft forever preferred_lft forever  
main@clientB:~$ ip route  
10.106.15.0/24 via 192.168.15.2 dev enp0s8 proto static metric 100  
172.22.15.0/24 via 192.168.15.1 dev enp0s8 proto static metric 100  
192.168.15.0/24 dev enp0s8 proto kernel scope link src 192.168.15.10 metric 100  
main@clientB:~$ ping -c 2 172.22.15.10  
PING 172.22.15.10 (172.22.15.10) 56(84) bytes of data.  
64 bytes from 172.22.15.10: icmp_seq=1 ttl=63 time=8.57 ms  
64 bytes from 172.22.15.10: icmp_seq=2 ttl=63 time=6.30 ms  
  
--- 172.22.15.10 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1002ms  
rtt min/avg/max/mdev = 6.303/7.435/8.568/1.132 ms  
main@clientB:~$ ping -c 2 10.106.15.10  
PING 10.106.15.10 (10.106.15.10) 56(84) bytes of data.  
64 bytes from 10.106.15.10: icmp_seq=1 ttl=63 time=2.53 ms  
64 bytes from 10.106.15.10: icmp_seq=2 ttl=63 time=1.44 ms  
  
--- 10.106.15.10 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1003ms  
rtt min/avg/max/mdev = 1.439/1.984/2.530/0.545 ms  
main@clientB:~$ ping -c 2 172.22.15.1  
PING 172.22.15.1 (172.22.15.1) 56(84) bytes of data.  
64 bytes from 172.22.15.1: icmp_seq=1 ttl=64 time=6.70 ms  
64 bytes from 172.22.15.1: icmp_seq=2 ttl=64 time=0.676 ms  
  
--- 172.22.15.1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 0.676/3.687/6.698/3.011 ms  
main@clientB:~$ ping -c 2 192.168.15.2  
PING 192.168.15.2 (192.168.15.2) 56(84) bytes of data.  
64 bytes from 192.168.15.2: icmp_seq=1 ttl=64 time=2.83 ms  
64 bytes from 192.168.15.2: icmp_seq=2 ttl=64 time=0.708 ms  
  
--- 192.168.15.2 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 0.708/1.768/2.829/1.060 ms
```

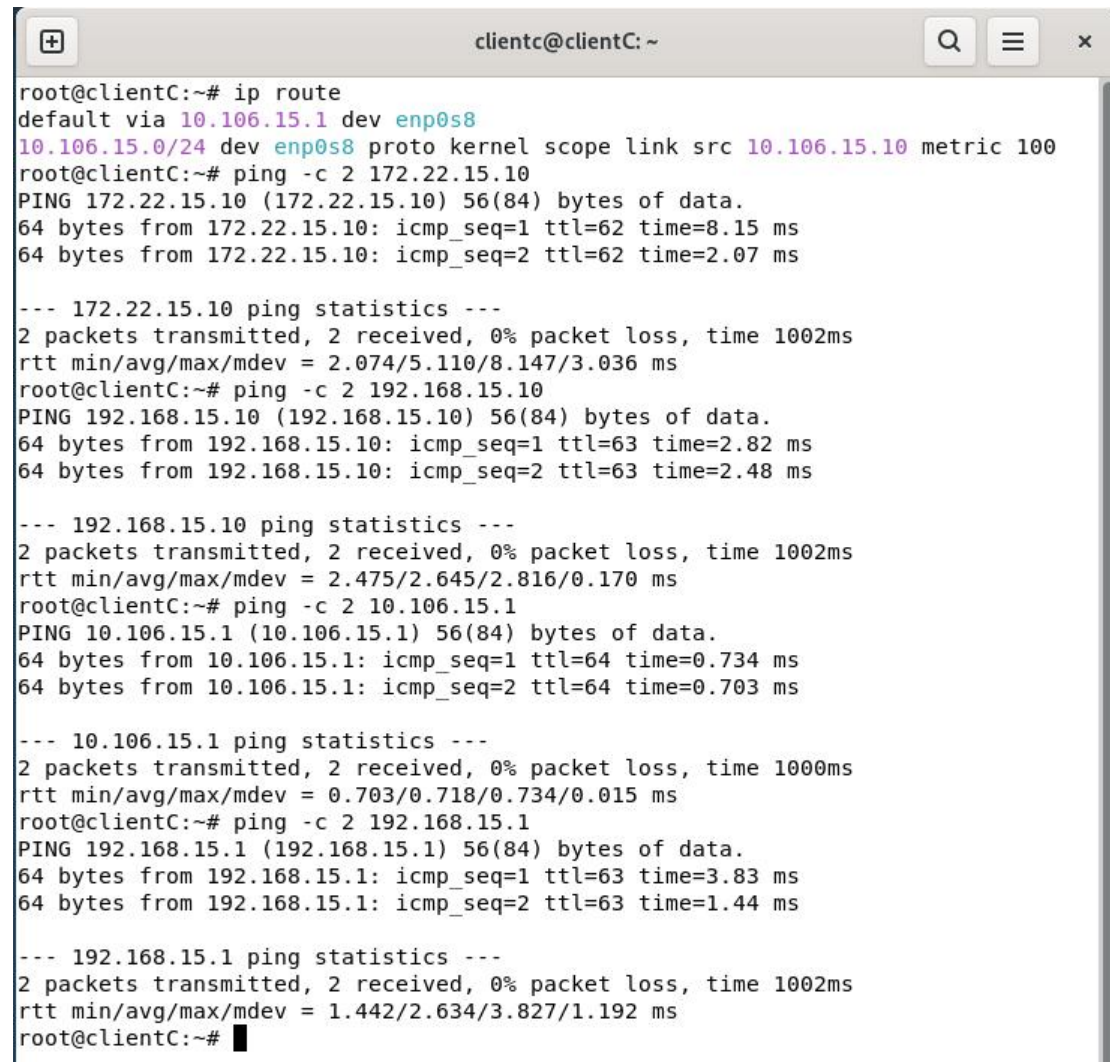
Для clientC

ping -c 2 172.22.15.10 # ClientA

ping -c 2 192.168.15.10 # ClientB

ping -c 2 10.106.15.1 # Router2 (local)

ping -c 2 192.168.15.1 # Router1 (through Router2)

A terminal window titled 'clientc@clientC: ~' with search, menu, and close buttons. It displays the output of several network commands. First, 'ip route' shows the default route via 10.106.15.1. Then, a series of ping tests are performed: to 172.22.15.10 (ClientA), 192.168.15.10 (ClientB), 10.106.15.1 (Router2), and 192.168.15.1 (Router1). Each ping test includes individual packet results and a summary statistics block showing 0% packet loss and round-trip times.

```
root@clientC:~# ip route
default via 10.106.15.1 dev enp0s8
10.106.15.0/24 dev enp0s8 proto kernel scope link src 10.106.15.10 metric 100
root@clientC:~# ping -c 2 172.22.15.10
PING 172.22.15.10 (172.22.15.10) 56(84) bytes of data.
64 bytes from 172.22.15.10: icmp_seq=1 ttl=62 time=8.15 ms
64 bytes from 172.22.15.10: icmp_seq=2 ttl=62 time=2.07 ms

--- 172.22.15.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.074/5.110/8.147/3.036 ms
root@clientC:~# ping -c 2 192.168.15.10
PING 192.168.15.10 (192.168.15.10) 56(84) bytes of data.
64 bytes from 192.168.15.10: icmp_seq=1 ttl=63 time=2.82 ms
64 bytes from 192.168.15.10: icmp_seq=2 ttl=63 time=2.48 ms

--- 192.168.15.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.475/2.645/2.816/0.170 ms
root@clientC:~# ping -c 2 10.106.15.1
PING 10.106.15.1 (10.106.15.1) 56(84) bytes of data.
64 bytes from 10.106.15.1: icmp_seq=1 ttl=64 time=0.734 ms
64 bytes from 10.106.15.1: icmp_seq=2 ttl=64 time=0.703 ms

--- 10.106.15.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1000ms
rtt min/avg/max/mdev = 0.703/0.718/0.734/0.015 ms
root@clientC:~# ping -c 2 192.168.15.1
PING 192.168.15.1 (192.168.15.1) 56(84) bytes of data.
64 bytes from 192.168.15.1: icmp_seq=1 ttl=63 time=3.83 ms
64 bytes from 192.168.15.1: icmp_seq=2 ttl=63 time=1.44 ms

--- 192.168.15.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.442/2.634/3.827/1.192 ms
root@clientC:~#
```

Для Router1

ping -c 2 192.168.15.2 # Router2

ping -c 2 10.106.15.10 # ClientC (through Router2)


```
alex@router1:~$ ip route
10.106.15.0/24 via 192.168.15.2 dev enp0s9 proto static metric 100
172.22.15.0/24 dev enp0s8 proto kernel scope link src 172.22.15.1 metric 101
192.168.15.0/24 dev enp0s9 proto kernel scope link src 192.168.15.1 metric 100
alex@router1:~$ ping -c 2 192.168.15.2
PING 192.168.15.2 (192.168.15.2) 56(84) bytes of data.
64 bytes from 192.168.15.2: icmp_seq=1 ttl=64 time=3.79 ms
64 bytes from 192.168.15.2: icmp_seq=2 ttl=64 time=7.44 ms

--- 192.168.15.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1006ms
rtt min/avg/max/mdev = 3.789/5.614/7.439/1.825 ms
alex@router1:~$ ping -c 2 10.106.15.10
PING 10.106.15.10 (10.106.15.10) 56(84) bytes of data.
64 bytes from 10.106.15.10: icmp_seq=1 ttl=63 time=4.00 ms
64 bytes from 10.106.15.10: icmp_seq=2 ttl=63 time=1.57 ms

--- 10.106.15.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.568/2.781/3.995/1.213 ms
alex@router1:~$
```

Для Router2

ping -c 2 192.168.15.1 # Router1

ping -c 2 172.22.15.10 # ClientA (through Router1)

```
john@router2:~$ ip route
10.106.15.0/24 dev enp0s9 proto kernel scope link src 10.106.15.1 metric 101
172.22.15.0/24 via 192.168.15.1 dev enp0s8
192.168.15.0/24 dev enp0s8 proto kernel scope link src 192.168.15.2 metric 100
john@router2:~$ ping -c 2 192.168.15.1
PING 192.168.15.1 (192.168.15.1) 56(84) bytes of data.
64 bytes from 192.168.15.1: icmp_seq=1 ttl=64 time=12.1 ms
64 bytes from 192.168.15.1: icmp_seq=2 ttl=64 time=4.98 ms

--- 192.168.15.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 4.975/8.537/12.100/3.562 ms
john@router2:~$ ping -c 2 172.22.15.10
PING 172.22.15.10 (172.22.15.10) 56(84) bytes of data.
64 bytes from 172.22.15.10: icmp_seq=1 ttl=63 time=2.50 ms
64 bytes from 172.22.15.10: icmp_seq=2 ttl=63 time=1.28 ms

--- 172.22.15.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1004ms
rtt min/avg/max/mdev = 1.282/1.891/2.500/0.609 ms
john@router2:~$
```

Выводы

В ходе данной лабораторной работы была успешно спроектирована и реализована маршрутизируемая сеть из трёх подсетей с использованием статической маршрутизации. Путём настройки IP-адресов, таблиц маршрутизации и шлюзов на двух маршрутизаторах и трех клиентских машинах была достигнута полная связность между всеми сегментами сети. Успешный обмен ping-запросами между всеми устройствами подтвердил корректность настроенных маршрутов. Практическая реализация закрепила теоретические знания о принципах маршрутизации, сегментации сетей и ключевой роли маршрутизаторов в обеспечении межсетевого взаимодействия. сетевой среде. Все компоненты системы работают согласованно и соответствуют требованиям задания.